

Model Questions

Module 3

Vector Semantic and Embedding

Lexical Semantics: basic concepts; Vector Semantics; Words and Vectors; Concepts of Cosine Similarity, TF-IDF, PMI; Word2Vec; Semantic Properties of Embedding.

Case Study: Study on Word2Vec model and relation between various words.

1. What is Vector Model? Write its advantages and disadvantages.
2. How can you calculate tf and idf in the vector model?
3. Consider a document represented by three terms {tornado, swirl, wind} with the raw tf 4, 1, and 1 respectively. In a collection of 100 documents, 15 documents contain the term tornado, 20 contains swirl, and 40 contain wind. Find the idf and weight of all the terms.
4. What is Vector Model? Write its advantages and disadvantages.
5. How can you find the similarity between doc and query in probabilistic principle using Bayes' rule?
6. Write the advantages and disadvantages of the probabilistic model
7. Fig. below shows a simplified subset of the word-word co-occurrence matrix for these four words compute from the Wikipedia corpus (Davies, 2015).

	computer	data	result	pie	sugar
Cherry	2	8	9	442	25
Strawberry	0	0	1	60	19
digital	1670	1683	85	5	4
information	3325	3982	378	5	13

Show spatial visualization of word vectors for digital and information In the two dimensions corresponding to the words data and computer,

8. Find cosine similarity in cherry with information, and digital with information from the figure below.

	pie	data	computer
Cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

9. Shows the joint probabilities and the PPMI values for four words in 5 contexts in the Wikipedia corpus as in fig. below.

	computer	data	result	pie	sugar
Cherry	2	8	9	442	25
Strawberry	0	0	1	60	19
digital	1670	1683	85	5	4
information	3325	3982	378	5	13

10. Advantages of Word2vec over tf-idf model.
11. Discuss briefly different types of static embeddings.
12. Explain semantic properties of embeddings with suitable examples.

Module-4

Sequence Labelling: Parts-of-speech Tagging

Parts-of-speech Tagging: basic concepts; Word Classes and Tagset; POS tagging using Hidden Markov Model; Working with a toy example; Concept of Named Entity Recognition.

Case Study: POS tagging in Indian languages.

1. Define a finite automaton that accepts the following language:
(aa)*(bb)*
2. Compute the minimum edit distance between tutor and tumour.
3. Compute the validity of the following statements:
 - a) Rule-based taggers are non-deterministic
 - b) Stochastic taggers are language independent
4. How can unknown words be handled in the tagging process?
5. What is NER and how does NER work? Explain with suitable example.
 - a) Assign correct POS tagging on each word then create an Emission and State transition probability matrix of the following Training Corpus based on HMM:
Book a Taxi. Park the Taxi.
The Taxi is in a park.
The book is in the Taxi.
 - b) Apply the Viterbi algorithm to the following test data using the above matrices:
“The park is a book”.